

Renatto Tommasi Hernández

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Current Residence Budapest, Hungary



PROFILE

Robotics Perception Engineer who takes autonomous mobile robots from raw sensor data to a reliable state estimate on real hardware, in the field. Owns the perception and localization stack of an autonomous ground vehicle in C++17 and Python (ROS 2): sensor bringup and intrinsic/extrinsic calibration, multi-camera visual odometry and SLAM, depth-image and point-cloud processing, free-space and obstacle estimation, and multi-sensor fusion through factor-graph optimization (GTSAM) on SE(3), optimized for NVIDIA Jetson edge hardware. Fluent in coordinate frames, rigid-body transforms (SO(3)/SE(3)) and 3D geometry, and comfortable bisecting a stack across sensor drivers, timing, networking, and geometric logic to find where reality and the math disagree. Builds the validation infrastructure that proves it works: Isaac Sim digital twins for simulation-first testing before every hardware deployment, large-scale ROS-bag field-data collection and replay for repeatable metrics, and 3D visualization in Foxglove and RViz to diagnose real-world failure modes. Hardware-native mechatronics background, first-author SLAM publication, and a decade in competitive robotics. English, German (C1), native Spanish.

STUDIES

- 08/2023 - 07/2025 M.Sc. Intelligent Field Robotics Systems**
Universitat de Girona and University of Zagreb
- Erasmus Mundus Joint Masters Scholar
 - **Relevant Coursework:** Hands-on Localization (10.0/10), Hands-on Perception (9.6/10), Robot Sensing, Perception and Actuation (9.6/10), Hands-on Planning (9.4/10), Machine Learning (8.4/10), Probabilistic Robotics, Multiview Geometry.
 - GPA: 8.75/10 · Thesis Grade: 9.1/10
Thesis: "Enhancing Indoor Mapping and Localization in Specular Rich Environments using Deep Learning and Sensor Fusion"
- 08/2018 - 06/2023 B.Sc. Mechatronics Engineering**
Tecnológico de Monterrey (ITESM) & Hochschule Mannheim (Double Degree)
- Double Degree at Hochschule Mannheim (DHIK programme, 17 months: coursework, internship, and thesis).
 - ITESM Scholarship for Excellence in Leadership · DHIK-DAAD Scholarship Holder
 - **Relevant Coursework:** Machine Vision & Autonomous Driving, Robotics, Industrial Robotics, Control Engineering, Signal & System Analysis, Mechatronics Design, Microcontrollers.
 - GPA: 89.04/100 · Thesis Grade: 1.0 (German scale)
Thesis: "Application of a force-torque sensor for the control of robotic drilling processes".

PROFESSIONAL EXPERIENCE

- 04/2026 - Current Senior Robotics Engineer**
European Knowledge Centre (Hungary)
- **ViroFighter Autonomous UGV — Perception & State Estimation:** Own the real-time perception and localization stack (ROS 2, C++17, Python), from raw sensor drivers to the poses and free-space estimates consumed by planning and control. Cut angular drift 95% by fusing multi-camera visual odometry (four fisheye cameras, SuperPoint features), RGB-D point clouds, and wheel odometry in a factor graph on SE(3) (GTSAM).
 - **Sensor Integration & Edge Deployment:** Bringup, intrinsic/extrinsic calibration, and time synchronization of the wide-FOV fisheye and RGB-D suite; optimized and deployed the stack to NVIDIA Jetson (Isaac ROS) under real-time constraints.
 - **Field Data & Validation Tooling:** Large-scale ROS-bag field-data collection and replay for metric-driven benchmarking, live logging to reproduce real-world failure modes, and 3D visualization of point clouds, trajectories, and frames in Foxglove and RViz. Docker CI across the stack.
 - **ROS 2 Fleet Migration & Simulation-First Validation:** Oversee the ROS 1 → ROS 2 migration of the robot fleet, and built the Isaac Sim environment used to develop and validate perception, localization, and navigation before every hardware deployment.

- **Applied Research Supervision:** Supervise two industrial M.Sc. theses — vision-language models for semantic localization, and behaviour cloning with actor-critic refinement in Nav2 — merging into one end-to-end vision-language-action architecture on a real mobile robot.

04/2026 - Current

ESA Payload Systems Engineer

European Knowledge Centre (Hungary)

- Systems Engineer for the Prime Contractor on an ESA ISS payload flying a point-of-care hematology analyser with minimal modification to the commercial device.
- Lead the consortium’s engineering effort with Airbus Defence and Space and report technical progress to ESA under ECSS standards toward PDR.
- Own payload design and product assurance across a multidisciplinary team of software, electrical, mechanical, and systems engineers.

07/2025 - 04/2026

AI & Robotics Engineer

European Knowledge Centre (Hungary)

- **Medical AI Team Lead (HUNOR ISS Mission):** Led the medical-AI team, in collaboration with Semmelweis University, that developed AI-assisted diagnostic and health-monitoring tools for the HUNOR programme that flew a Hungarian astronaut to the ISS — an activity and fitness tracker, a cuffless blood-pressure estimator, and a denoising filter for the Astroskin biometric sensor suit, all trained and validated on noisy real-world wearable sensor data.
- **ViroFighter Debugging & System Testing (ROS 1):** Owned debugging and system-level testing of the ViroFighter autonomous UGV on its ROS 1 Melodic stack — reproducing field failures and isolating faults across sensor drivers, timing and network layers, localization, and geometric/calibration logic to determine whether a bug lived in the hardware, the data, or the math.
- **Computer-Vision Quality Assurance (Mylan):** Built an automated optical inspection station in C# for a pharmaceutical packaging line, detecting print defects on the backing foil of medicine blister packs at end of line — image acquisition, defect detection, and pass/fail decisioning integrated into the production QA workflow.
- **Emerging Perception Technology:** Evaluated and shipped a DINOv3 foundation-model anomaly detector to production (adopted by Airbus Geo, 50% less data-cleaning time), assessing accuracy, runtime cost, and production readiness against classical baselines.

02/2025 - 06/2025

Intern — Robotic Perception & Localization

European Knowledge Centre (Hungary)

- **Free-Space Estimation & Sensor Noise Rejection:** Developed a Support Vector Machine (SVM) classifier to reject reflection artifacts from Intel RealSense depth data, improving free-space and obstacle estimation in perceptually challenging (specular) environments. Supported calibration and time synchronization between the four wide-FOV fisheye cameras and the RGB-D cameras.
- Prototyped the multi-camera Visual Odometry pipeline (four fisheye cameras, SuperPoint deep-learning features, GTSAM graph optimization) that matured into the ViroFighter localization stack, cutting angular drift by 95%.

04/2022 - 03/2023

Intern & Bachelor Thesis Student — Robotic Automation

Robert Bosch Power Tools GmbH (Germany)

- **Manipulator Work Cell (UR16e):** Designed and built the electromechanical test bench for a UR16e-based robotic drilling system destined for an autonomous drilling AGV — interfacing with the arm’s control software to command end-effector poses across the workspace and closing the loop with real-time sensor feedback on physical hardware.
- **Sensing & Low-Level Integration:** Developed a redundant force-torque sensing system (F/T sensor, accelerometer, motor current) for real-time characterization of contact forces and torques across drilling poses, including data acquisition, synchronization, and analysis of the multi-sensor streams.

PROJECTS

2024

Aerial Robotics — Vision-Based State Estimation

- Deployed ORB-SLAM2 for robust, purely vision-based state estimation and localization on a Tello UAV in GPS-denied environments.

2023

Multi-Object Tracking using Kalman Filter on Event-Based Data

- Implemented a high-frequency Kalman Filter for trajectory and velocity estimation from asynchronous Dynamic Vision Sensor (DVS) streams, engineering the data pipeline around continuous, microsecond-resolution events rather than synchronous frames for low-latency tracking.

- 2023
- Autonomous Navigation onboard a TurtleBot 3

 - Feature-based Graph SLAM localization using ArUco marker landmarks and the GTSAM library
 - Behavior-tree exploration with frontier-based RRT* planning under kinematic constraints, pure-pursuit PID path tracking, and a task-priority forward-kinematics manipulator controller for object intervention.

2017 - 2023

FIRST Robotics Competition (FRC)

- Founder, former captain and mentor of the Prepa Tec Daedalus 7102 team; robot inspector certifying 50 industrial-category robots against FRC event regulations.

PUBLICATIONS

R. Tommasi Hernández, Dr. G. Vasilevic and Prof. Z. Istenes. "Robust Multi-Camera Visual Odometry in Specular Indoor Environments Using Deep Learning," in *Proc. IntRob26*, 2026. **(Accepted for publication)**

R. Tommasi Hernández, Dr. G. Bohus, Dr. J. Nagy-Bozsoky, P. Cordero, Z. Horvath, Dr. K.V. Nagy and Prof. B. Merkely, "Predicting Hemodynamic Shifts in Microgravity: A Deep Learning Approach to Cuffless Blood Pressure Estimation," in *Proc. IAC2026*, 2026. **(Accepted for publication)**

LANGUAGES

Spanish: Native • **English:** C1 (TOEFL iBT) • **German:** C1 (Goethe Institut)

SKILLS

Perception & Computer Vision	SLAM & Visual Odometry (multi-camera, factor-graph optimization), Localization & State Estimation, Multi-Sensor Fusion, Monocular / Fisheye / RGB-D Depth Cameras, Point-Cloud & Depth-Image Processing, Free-Space & Obstacle Estimation, Object Detection & Tracking, Automated Optical Inspection, Deep-Learning Features (SuperPoint) & Vision Foundation Models (DINOv3), Kalman Filtering.
3D Geometry & Robotics	Rigid-Body Transforms (SO(3)/SE(3)), Rotations & Quaternions, Coordinate Frames & Transform Trees (TF2), Camera Intrinsics/Extrinsics & Hand-Eye Calibration, Motion Planning & Obstacle Avoidance (Nav2), Probabilistic Robotics.
Programming & Libraries	Modern C++ (14/17), Python, ROS 2 (incl. ROS 1 → ROS 2 migration), Linux, OpenCV, PCL / Open3D, Eigen, GTSAM, NumPy / SciPy, PyTorch, TensorRT, CUDA.
Systems, Tooling & Hardware	Sensor Bringup, Calibration & Time Synchronization, Embedded / Edge Deployment & Optimization (NVIDIA Jetson, Isaac ROS), Simulation-First Validation (Isaac Sim), Large-Scale ROS-Bag Field-Data Collection & Replay, Logging, Metrics & Evaluation Tooling, 3D Data Visualization (Foxglove, RViz, Open3D), AI-Assisted Development & Agents (implementation, debugging, code review), Real-Time Systems, Docker, Git, CI/CD.
Soft Skills	Cross-Functional Collaboration (Perception / Controls / Embedded / Hardware / Systems), End-to-End Ownership, Field Testing & Debugging on Physical Robots, Team Leadership & Mentoring.